

Research Notes

Research question: How can one static HTML page be delivered globally over HTTPS while its S3 origin remains private and the environment remains reproducible?

Private static origin

Finding: S3 stores index.html without operating servers. Public Access Block and BucketOwnerEnforced remove public ACL-based delivery.

Evidence basis: AWS S3 data protection, Public Access Block, and Object Ownership documentation.

Global HTTPS delivery

Finding: CloudFront supplies a globally distributed HTTPS endpoint, managed edge caching, compression, and elastic request handling.

Evidence basis: AWS CloudFront introduction and managed cache policy documentation.

Origin restriction

Finding: Origin Access Control signs CloudFront-to-S3 requests with SigV4. The bucket policy limits s3:GetObject to the created distribution SourceArn.

Evidence basis: AWS CloudFront guidance for restricting an S3 origin.

Reproducibility

Finding: Terraform represents the bucket, security controls, object, OAC, distribution, and policy as reviewable configuration.

Evidence basis: HashiCorp Terraform introduction, CLI workflow, and AWS provider documentation.

Alternatives

Finding: EC2 and a load balancer add patching, capacity, and operating-system responsibilities that a single static file does not require. Lambda, databases, APIs, and containers have no workload in scope.

Evidence basis: Reasoned comparison against the functional requirement: serve one static HTML object.

Verification

Finding: Positive tests cover HTTPS, redirect, deployment, cache hit, and no-change plan. Negative tests confirm direct S3 denial and all four Public Access Block values.

Evidence basis: Genuine terminal and browser evidence captured during deployment.

Research conclusion

The smallest suitable architecture is CloudFront with OAC in front of a private encrypted and versioned S3 bucket, provisioned by Terraform. This directly addresses availability, global delivery, automatic scaling, HTTPS, private origin storage, and reproducibility without introducing compute services that the page cannot use.